

Class - 6

Complex numbers

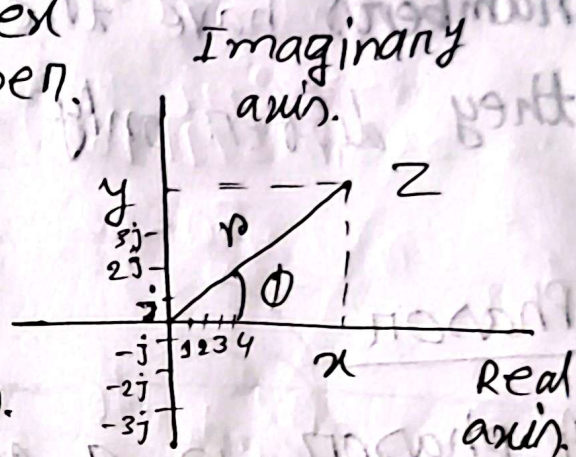
$$z = x + jy \text{ [Rectangular form]}$$

$$z = r \angle \phi \text{ [Polar form]}$$

$$z = r e^{j\phi} \text{ [Exponential form]}$$

$$= r \cos \phi + j r \sin \phi$$

Complex number.



$$r = \sqrt{x^2 + y^2}, \quad y = r \sin \phi$$

$$x = r \cos \phi, \quad \phi = \tan^{-1} \frac{y}{x}$$

transform.

$$\star. \quad z_1 = x_1 + jy_1 = r_1 \angle \phi_1$$

$$z_2 = x_2 + jy_2 = r_2 \angle \phi_2$$

$$i = \sqrt{-1}$$

$$j = \sqrt{-1}$$

Addition: $z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$

Subtraction: $z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$

Multiplication: $z_1 \cdot z_2 = r_1 r_2 \angle \phi_1 + \phi_2$

Division: $\frac{z_1}{z_2} = \frac{r_1 \angle \phi_1}{r_2 \angle \phi_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2$

Reciprocal: $\frac{1}{z} = \frac{1}{r} \angle -\phi$

❖ Addition and subtraction can not be performed in polar form unless the complex numbers have the same angle or unless they differ only by multiples of 180°

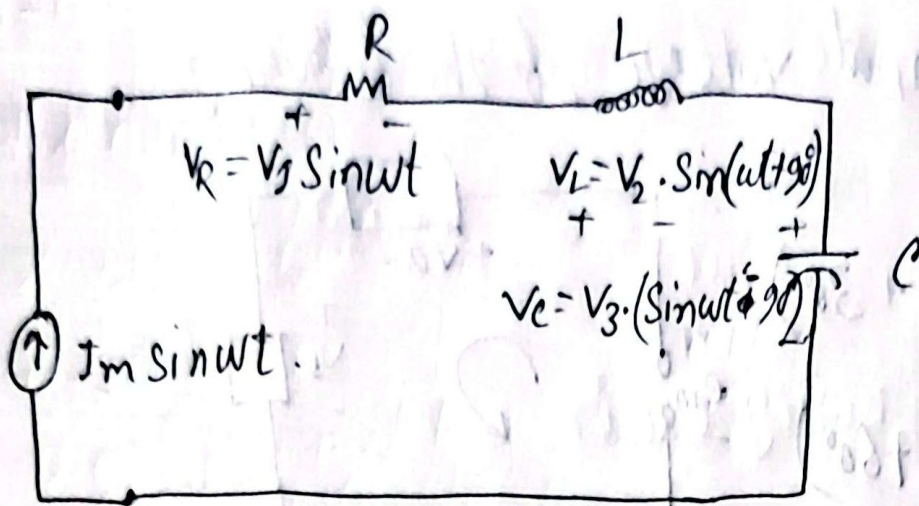
Phasor

A phasor is a complex number that represents the amplitude and phase of a sinusoid.

$$V(t) = V_m \cos(\omega t + \phi) \quad \Leftrightarrow \quad V = \frac{V_m}{\sqrt{2}} \angle \phi$$

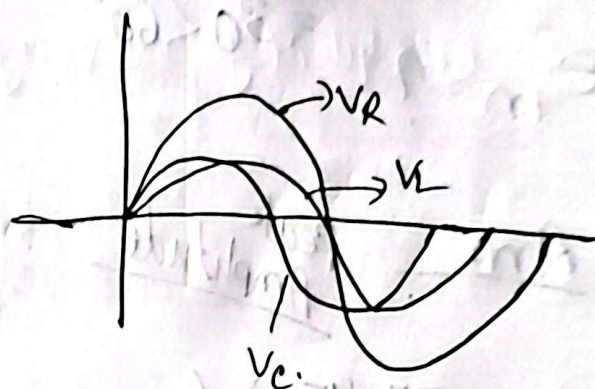
Time Domain Representation. $V_m \sin(\omega t + \phi)$

Phasor Domain Representation. $V_{rms} \angle \phi$



$$V_T = V_R + V_L + V_C$$

$$= V_1 \sin wt + V_2 \sin(wt + 90^\circ) + V_3 \sin(wt - 90^\circ)$$



Time

Phasor

a) $\sqrt{2} (50) \sin wt$

$50 \angle 0^\circ$

b) $69.6 \sin(wt + 72^\circ)$

$(0.707) / (69.6) \angle 72^\circ = 49.2 \angle 72^\circ$

c) $45 \cos wt$

$(0.707) (45) \angle 90^\circ = 31.82 \angle 90^\circ$

Phasor

Time

a) $I = 10 \angle 30^\circ$

$i = \sqrt{2} 10 \sin(2\pi 60t + 30^\circ)$

and $i = 14.14 \sin(377t + 30^\circ)$

b) $V = 115 \angle -70^\circ$

$v = \sqrt{2} (115) \sin(377t - 70^\circ)$

and $v = 162.6 \sin(377t - 70^\circ)$

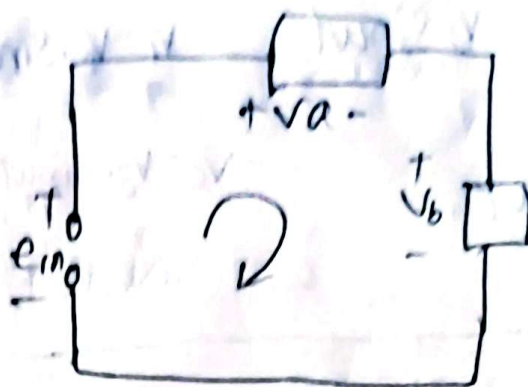
Q) Find the input voltage of the following circuit.

$$V_a = 50 \sin(377t + 30^\circ)$$

$$= 50 \angle 30^\circ$$

$$V_b = 30 \sin(377t + 60^\circ)$$

$$e_{in} = ? = 30 \angle 60^\circ$$



$$-e_{in} + V_a + V_b = 0$$

$$e_{in} = V_a + V_b$$

$$e_{in} = 77.45 \sin(377t + 41.167^\circ)$$

$$= 50 \angle 30^\circ + 30 \angle 60^\circ$$

$$= 77.44 \angle 41.167^\circ$$

$$= 58.30 + 50.98j$$

rms

$$V_a = \frac{50}{\sqrt{2}} \angle 30^\circ, V_b = \frac{30}{\sqrt{2}} \angle 60^\circ$$

$$e_{in} = V_a + V_b$$

$$= \frac{50}{\sqrt{2}} \angle 30^\circ + \frac{30}{\sqrt{2}} \angle 60^\circ$$

$$= 54.76 \angle 41.16^\circ$$

$$(e_{in}^{(+)}) = (\sqrt{2})(54.76) \sin(377t + 41.16^\circ)$$

$$= 77.44 \sin(377t + 41.6^\circ) \text{ volt}$$

Ans